Check for updates

Basic & Clinical Pharmacology & Toxicology

WILEY

ORIGINAL ARTICLE OPEN ACCESS

Regional Variation in ADHD Treatment and Diagnosis Among Children and Adolescents in Denmark

Heidi Sonne[1] | Anton Pottegård[1] | Anna Sofie Kjærgaard Hansen[2,3] | Mickael Bech[4] | Helene Kildegaard[1,5]

[1]Clinical Pharmacology, Pharmacy and Environmental Medicine, Department of Public Health, University of Southern Denmark, Odense, Denmark | [2]Psychiatry, North Denmark Region, Aalborg University Hospital, Aalborg, Denmark | [3]Department of Clinical Medicine, Aalborg University, Aalborg, Denmark | [4]Department of Political Science and Public Management, University of Southern Denmark, Odense, Denmark | [5]Mary Elizabeth's Hospital, University Hospital Copenhagen, Rigshospitalet, Denmark

Correspondence: Helene Kildegaard (hckildegaard@health.sdu.dk)

Received: 11 May 2026 | **Revised:** 17 August 2026 | **Accepted:** 18 August 2026

Keywords: adolescent psychiatry | attention deficit disorder with hyperactivity | child psychiatry | Denmark | geographic variation

ABSTRACT

Geographic variation in ADHD treatment and diagnosis among children and adolescents is well documented, yet the system-level factors driving such differences remain poorly understood. We examined nationwide patterns of ADHD medication use and diagnoses in Denmark in 2024 and assessed whether municipal socioeconomy, regional hospital waiting times or private child psychiatrist capacity contributed to geographic variation. In this register-based cross-sectional study, we included all Danish residents aged 4–17 years using data from national registers. Pharmacological treatment was defined as at least one redeemed prescription for ADHD drugs, and diagnosis was defined as at least one hospital contact with ICD-10 codes F90.x/F98.8. Prevalence was calculated overall and stratified by sex and age, with associations explored using Spearman's rank correlations and regression analyses. ADHD medication prevalence varied sixfold across municipalities (9.6–58 per 1000), with parallel variation in diagnoses (15–74 per 1000) and the largest differences observed among adolescents. Municipal socioeconomic status was weakly and inversely correlated with medication use ($\rho = -0.20$, $p = 0.049$), but not diagnoses. Neither waiting times nor specialist capacity was significantly associated with either outcome. Marked geographic variation in ADHD medication use and diagnoses persists in Denmark despite a uniform healthcare framework, and appears largely unexplained by the system-level factors examined.

1 | Introduction

The prevalence of attention-deficit/hyperactivity disorder (ADHD) diagnoses has increased substantially over recent decades [1]. International studies have documented marked regional variation in both the diagnosis and treatment of ADHD, including evidence from Norway, Sweden and the United States [2–4]. These differences are only partly explained by underlying patient needs [2] and may also reflect supply- and demand-side factors, including access to services and local medical practices [5, 6].

Geographic variation in healthcare utilisation has been well documented for decades, often referred to as 'small area variation' [5]. Such differences are typically regarded as unwarranted when they cannot be attributed to evidence-based guidelines or population health needs [7]. Similar concerns apply to ADHD, where local practices and system-level factors may drive differential access to diagnostic and treatment services.

In Denmark, approximately 9% of children receive an ADHD diagnosis before age 18 [1]. Earlier Danish research has found substantial regional variation in incidence, with lower rates in the Region of Southern Denmark, particularly on Funen, and higher rates in the Zealand Region [8]. These findings point to unwarranted variation in care delivery and highlight potential inequalities in access to diagnosis and treatment.

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2026 The Author(s). *Basic & Clinical Pharmacology & Toxicology* published by John Wiley & Sons Ltd on behalf of Nordic Association for the Publication of BCPT (former Nordic Pharmacological Society).

Plain Language Summary

We examined how often children and adolescents in Denmark were diagnosed with ADHD or received ADHD medication across municipalities in 2024. Despite a tax-funded healthcare system with equal access, we found large geographic differences. Some municipalities had more than six times as many children receiving ADHD medication as others, with the largest differences seen among adolescents. Municipal socioeconomic indicators were only weakly associated with these patterns, and differences could not be explained by waiting times or specialist availability. The findings suggest that factors beyond medical need may influence access to ADHD diagnosis and treatment in Denmark.

This study provides updated nationwide estimates of geographic variation in ADHD treatment in Denmark in 2024. We further examine system-level factors that may drive these differences, to assess whether regional variation reflects true population needs or unwarranted variation in care. By doing so, we aim to generate evidence that can inform strategies to promote more equitable access to ADHD treatment across the country.

2 | Methods

We conducted a nationwide register-based study to estimate the prevalence of ADHD diagnosis and medication use among Danish children and adolescents, and to examine selected regional and municipal indicators potentially associated with variation. Data were obtained from the Danish National Patient Register [9], the Danish National Prescription Registry [10], the Danish Civil Registration System [11], the Ministry of the Interior and Health [12] and the Danish Health Data Authority [13–15].

2.1 | Setting

Denmark has a universal, tax-funded healthcare system in which all residents are entitled to publicly financed healthcare. Access to diagnostic assessments, consultations and hospital-based services is provided free of charge, while a fixed percentage copayment applies to outpatient prescription drugs [16]. The system is organised into five regions responsible for managing hospital services, general practitioners and private specialist care [17]: The Capital Region of Denmark, Region Zealand, the Region of Southern Denmark, the Central Denmark Region and the North Denmark Region [16, 18], which are shown in Figure S1. These regions are subdivided into 98 municipalities, most of which have at least 20 000 inhabitants [18].

Child and adolescent psychiatry in Denmark is organised as a specialised regional service [17]. Access requires referral from a general practitioner, the municipality or another authorised healthcare professional, after which referrals are assessed by the regional child and adolescent psychiatry services [19, 20]. Once a referral is accepted, the patient undergoes a standardised diagnostic assessment, followed by a treatment plan that may include pharmacological treatment. Depending on symptom severity, complexity and waiting time, patients may be referred to private practicing child and adolescent psychiatry clinics working under agreements with the public health insurance system [21]. In 2023, approximately 5000 patients were treated by private child and adolescent psychiatrists working under the public reimbursement system, compared with about 48 700 patients in hospital-based child and adolescent psychiatry [21]. An unknown number of private child and adolescent psychiatrists work outside the public reimbursement system, requiring full out-of-pocket payment by families [21].

Diagnostic information from private practices is not systematically reported to the Danish National Patient Register, while all prescriptions redeemed at pharmacies are mandatorily recorded in the Danish National Prescription Registry [10], enabling more complete monitoring of medication use compared with diagnostic data.

2.2 | Study Population

The study population consisted of all individuals aged 4–17 years residing in Denmark between 1 January and 31 December 2024. The population was obtained from the Danish Civil Registration System along with key demographic information, including date of birth, sex and residential region and municipality. The age range was chosen from 4 years, as ADHD symptoms can be difficult to distinguish from normative behaviour before this age [22], and diagnosis before age 4 is rare [23].

2.3 | Analyses

Prevalence estimates were calculated overall and stratified by sex, age groups (4–6, 7–12 and 13–17 years), region and municipality. Age groups were chosen to align with key stages in the Danish educational system: kindergarten and preschool (4–6 years), primary and middle school (7–12 years) and secondary school to early upper secondary education (13–17 years). For analyses stratified by municipality, residence as of 31 December 2024, was used to assign individuals. The denominator was estimated as the number of individuals within the age range in each region and municipality alive on 1 July 2024 (Table S1).

Users of ADHD medication were defined as individuals who redeemed at least one prescription for methylphenidate (Anatomic Therapeutic Chemical [ATC] code: N06BA04, 62% of all defined daily doses filled in 2024 for age 0–17) [24], lisdexamphetamine (N06BA12; 29%), atomoxetine (N06BA09; 6.7%), dexamphetamine (N06BA02; 1.4%), or guanfacine (C02AC02; 1.1%) between 1 January and 31 December 2024, identified using the Danish National Prescription Registry [10].

The prevalence of ADHD diagnoses was defined as the proportion of individuals in the study population who had received a diagnosis before 31 December 2024, using all available lookback. An ADHD diagnosis was defined as at least one hospital contact, either inpatient or outpatient, recorded in the Danish National Patient Register with a diagnosis according to the

17427843, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/bcpt.70298 by Stony Brook University Libraries, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

International Classification of Diseases, 10th revision (ICD-10), codes F90.x or F98.8 [9, 25].

To explore potential drivers of geographic variation, we included three types of regional and municipality-level indicators, capturing both demand- (1) and supply-side factors (2 + 3): (1) The municipal socioeconomic index, obtained from the Ministry of the Interior and Health, which estimates each municipality's relative expenditure needs based on a range of indicators, including the proportion of adults not in employment and the number of residents with psychiatric conditions [12]. Values above 1 indicate higher-than-average expenditure needs, while values below 1 indicate lower-than-average needs [12]. (2) Waiting time in the hospital sector, obtained from the Danish Health Data Authority. We examined two waiting time indicators: (a) the time from referral to completed diagnostic assessment, defined as the number of calendar days from referral to completed assessment [15] and (b) the time from the completed diagnostic assessment to treatment initiation, or, if workup occurred outside the hospital system, from the receipt of referral to treatment start [13]. For both indicators, quarterly regional estimates for the third and fourth quarters of 2024 were averaged. (3) The number of private practicing child and adolescent psychiatrists working under the public reimbursement system, obtained from the Danish Health Data Authority and expressed as full-time equivalent capacities (FTEs) per 100 000 inhabitants, where a full-time practice counted as one FTE and part-time practices typically counted as one-third FTE [14].

Associations between municipal and regional indicators and the prevalence of ADHD diagnoses and medication use were assessed using Spearman's rank correlation coefficients (ρ). For the municipal socioeconomic index, we additionally fitted population-weighted linear regression models with prevalence proportion as the outcome. For the regional indicators (waiting time and psychiatrist capacity), associations were further examined using simple linear regressions based on regional prevalence estimates.

All analyses were performed using R version 4.4.3 and validated internally by a data manager not otherwise involved in the study.

The study was conducted in accordance with the Basic & Clinical Pharmacology & Toxicology policy for experimental and clinical studies [26].

3 | Results

In 2024, the prevalence of ADHD medication use among Danish children aged 4–17 years varied substantially across municipalities, ranging from 9.6 to 17 per 1000 in municipalities on Funen in the Region of Southern Denmark to 30 to 58 per 1000 in municipalities in the North Denmark Region and the Central Denmark Region (Figure 1A; Table S2). ADHD diagnoses followed a similar pattern, with the lowest prevalence (15–26 per 1000) in municipalities on Funen in the Region of Southern Denmark and the highest (32–74 per 1000) in municipalities in the North Denmark Region and the Central Denmark Region (Figure 1B).

When stratified by age, variation in medication use was lowest among children aged 4–6 years (1.2–4.7 per 1000 across municipalities with reportable numbers), higher among children aged 7–12 years (8.9–62 per 1000) and highest among adolescents aged 13–17 years (12–94 per 1000) (Figure S2; Table S3). ADHD diagnoses showed the same age-dependent pattern, with the widest geographic variation among adolescents (22–113 per 1000), somewhat narrower variation in children aged 7–12 years (15–74 per 1000) and low rates in children aged 4–6 years (3.1–10 per 1000) (Figure S3; Table S4).

When stratified by sex, ADHD medication use was consistently higher among boys than girls. For girls, municipal prevalence ranged from 7.6 to 46 per 1000, while the range for boys was 10–73 per 1000 (Figure S4; Table S5). Diagnostic prevalence showed a similar sex difference, ranging from 9.3 to 48 per 1000 among girls and 21–100 per 1000 among boys (Figure S5; Table S6).

Socioeconomic index values varied across municipalities with no clear regional pattern (Figure 1C; Table S7). ADHD medication use was slightly higher in municipalities with lower relative expenditure needs ($\rho = -0.20$, $p = 0.049$), whereas no significant association was found for ADHD diagnoses ($\rho = -0.042$, $p = 0.68$) (Table S3). In sex-stratified analyses, the inverse correlation between socioeconomic index and ADHD medication use was numerically stronger in girls ($\rho = -0.29$, $p = 0.005$) than in boys ($\rho = -0.15$, $p = 0.15$). Weighted linear regression yielded similar findings, showing a statistically significant inverse association between socioeconomic index and ADHD medication use ($\beta = -0.013$, $p = 0.027$), but no association with ADHD diagnoses ($\beta = -0.0041$, $p = 0.52$) (Table S9). However, there was no evidence that the association between socioeconomic index and either ADHD medication use or ADHD diagnoses differed by sex (interaction $p = 0.86$ and $p = 0.85$, respectively) (Table S10).

Regional waiting times varied considerably. The average waiting time from referral to completed diagnostic assessment ranged from 108 days in the Capital Region of Denmark to 223.5 days in the North Denmark Region, while the average waiting time from completed diagnostic assessment to treatment initiation ranged from 27.5 to 54 days (Figure 1E, Figure 1F; Table S3). However, neither waiting time measure was significantly associated with ADHD medication use or ADHD diagnoses (Table S8; Table S9).

Capacity in private child and adolescent psychiatry working under the public reimbursement system ranged from 1.9 per 100 000 in the Central Denmark Region to 3.5 per 100 000 in the North Denmark Region, but was likewise not significantly associated with medication use ($\rho = 0.40$, $p = 0.50$) or ADHD diagnoses ($\rho = 0.30$, $p = 0.62$) (Figure 1D; Table S8).

4 | Discussion

Our nationwide register-based study found marked geographic variation in ADHD diagnoses and medication use among Danish children in 2024, with prevalence proportions differing more than sixfold across municipalities. Municipal socioeconomic status, private specialist capacity and public hospital waiting times explained very little of these differences. Importantly,

17427843, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/bcpt.70298 by Stony Brook University Libraries, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

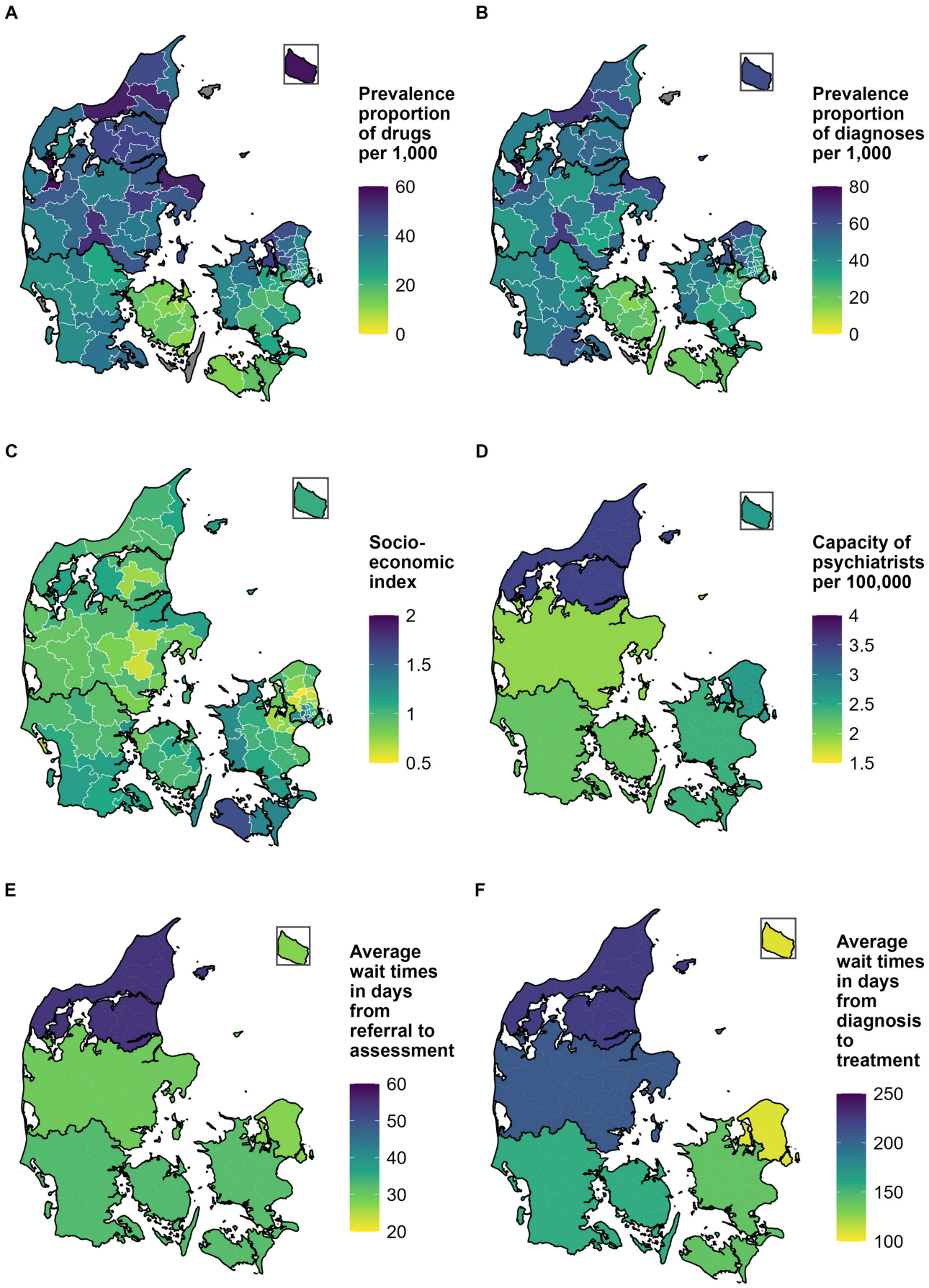


FIGURE 1 | Legend on next page.

17427843, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/bcpt.70298 by Stony Brook University Libraries, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

FIGURE 1 | Prevalence proportion of ADHD medication use and ADHD diagnoses among children and adolescents aged 4–17 years in Denmark, 2024, as well as municipal and regional indicators. Corresponding numerical values are provided in Tables S2 and S7. Grey areas indicate municipalities with non-reportable counts ($N < 10$). Municipal prevalence of ADHD medication use in 2024 defined as ≥1 redeemed prescription for methylphenidate (N06BA04), atomoxetine (N06BA09), dexamphetamine (N06BA02), lisdexamphetamine (N06BA12) or guanfacine (C02AC02), per 1000 inhabitants aged 4–17 years. Municipal prevalence of ADHD diagnoses in 2024 defined as ≥1 inpatient or outpatient hospital contact with ICD-10 F90.x or F98.8, per 1000 inhabitants aged 4–17 years. Municipal socioeconomic index (values >1 indicate higher-than-average needs). Regional capacity of private practicing psychiatrists working under the public reimbursement system, expressed as full-time equivalents per 100 000 inhabitants aged 4–17 years. Regional waiting times as the average of mean days from referral to diagnostic assessment in the 3rd and 4th quarter of 2024. Regional waiting times as the average of mean days from completed diagnostic workup or referral to treatment initiation in the 3rd and 4th quarter of 2024. (A) Municipal prevalence of ADHD medication use in 2024 defined as ≥1 redeemed prescription for methylphenidate (N06BA04), atomoxetine (N06BA09), dexamphetamine (N06BA02), lisdexamphetamine (N06BA12) or guanfacine (C02AC02), per 1000 inhabitants aged 4–17 years. (B) Municipal prevalence of ADHD diagnoses in 2024 defined as ≥1 inpatient or outpatient hospital contact with ICD-10 F90.x or F98.8, per 1000 inhabitants aged 4–17 years. (C) Municipal socioeconomic index (values >1 indicate higher-than-average needs). (D) Regional capacity of private practicing psychiatrists working under the public reimbursement system, expressed as full-time equivalents per 100 000 inhabitants aged 4–17 years. (E) Regional waiting times as the average of mean days from referral to diagnostic assessment in the 3rd and 4th quarter of 2024. (F) Regional waiting times as the average of mean days from completed diagnostic workup or referral to treatment initiation in the 3rd and 4th quarter of 2024.

such variation should not be interpreted as low prevalence being inherently preferable or high prevalence being undesirable. Rather, the concern is that the observed magnitude of variation appears too great given the uniform health system and lack of clear explanatory factors in our data.

Similar variations in ADHD diagnoses have been reported previously in Denmark, with high diagnostic rates in Region Zealand and low rates in the Region of Southern Denmark, especially on Funen [8]. Comparable findings have been observed in Norway, where geographic differences in diagnostic rates greatly exceeded variation in underlying ADHD symptom levels, indicating that morbidity alone cannot account for geographic patterns [3]. Likewise, regional variability in ADHD within a universal health system has been documented in Spain [27], suggesting that structural uniformity does not eliminate local-level differences.

In our study, ADHD diagnoses and medication use showed broadly similar geographic patterns, even though they reflect different observation windows: cumulative diagnosed prevalence for diagnoses and one-year treatment prevalence for medication use.

Although previous studies have reported higher ADHD prevalence in disadvantaged areas [28, 29], and more consistent pharmacotherapy among children from wealthier families [30], we observed no clear socioeconomic gradient in ADHD diagnoses. Medication use, however, showed a weak negative correlation with the socioeconomic index, suggesting slightly higher use in municipalities with lower relative expenditure needs. Earlier Danish studies also point in different directions: one study found considerable geographic clustering of ADHD incidence and medication use, but no significant associations with socioeconomic or municipal-level factors [31], while another study showed that regional prescribing differences were particularly pronounced among children facing social adversity, reflecting strong social gradients in some regions but not in others [32]. Norwegian data show that socioeconomic indicators are more strongly associated with service contact rates than with ADHD prevalence once children access mental health services [2]. They suggest that inequalities act mainly at the access stage, not at the diagnostic stage. Conceptual frameworks of healthcare access further highlight that such differences may occur at multiple stages of the care pathway, from the perception of needs to health care seeking, reaching and engagement, and not only at the point of diagnosis [33].

Specialist availability and waiting times may also contribute to geographic variation in access to child and adolescent psychiatry and thereby prevalence of ADHD. However, we found no statistically significant associations between either specialist capacity or regional waiting times and the prevalence of ADHD medication use or diagnoses. Results from the United States show that higher numbers of relevant physicians are associated with greater ADHD diagnosis and medication rates [34]. This suggests that supply-sensitive care may be a partial driver: higher availability of specialist services may increase detection and diagnosis, even when demand is constant [7]. The absence of clear associations in our study may reflect that these regional indicators do not fully capture local differences in access to care, or the limited statistical power of analyses based on five regions. Additional drivers such as parental demand, municipal referral pathways or local diagnostic culture may play an important role. Research from Norway has shown inter-clinician variation in diagnostic thresholds [35], and a similar factor could plausibly influence Danish practice despite uniform national guidelines. Evidence from Denmark further indicates that specialist physicians differ considerably in their prescribing behaviour across hospitals, and that such variation directly affects the probability that children receive pharmacological treatment [36]. Such preference-sensitive care, driven by differing professional judgements, remains a likely contributor [7].

Educational system incentives may also play a role. In some municipalities, access to special educational support requires a formal diagnosis, which can create incentives for parents, schools and professionals to seek diagnostic assessments. A recent qualitative study highlights how ADHD diagnoses are used institutionally in some Danish primary schools as a prerequisite for categorising children with emotional and behavioural difficulties and for allocating support services [37]. Such mechanisms could plausibly contribute to municipal differences if the

17427843, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/bcpt.70298 by Stony Brook University Libraries, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

reliance on diagnostic categorisation varies across local school systems.

Community-level dynamics may further shape ADHD prevalence. In autism, proximity to diagnosed children increases the likelihood of receiving a diagnosis, likely through heightened awareness and parent-to-parent information exchange [38]. Similar processes may influence ADHD patterns.

4.1 | Future Research

Beyond the factors examined in this study, other demand- and supply-side mechanisms could also influence regional variation in ADHD treatment. On the demand side, parental health-seeking behaviour, school-level incentives and local awareness networks may shape both referral patterns and treatment uptake. On the supply side, differences in municipal Educational Psychological Counselling Services (PPR), which often constitute the first step in the referral pathway to child and adolescent psychiatry, as well as distance to care, the distribution of private practising and public hospital-based child and adolescent psychiatrists, regional prioritisation of resources and varying professional norms, including potential differences between private and hospital-based diagnostic practices, could contribute. Future research should therefore combine register data with qualitative and survey-based approaches to capture these dimensions and assess how such mechanisms interact with structural factors to generate the observed variation.

4.2 | Strengths and Limitations

Key strengths include the use of nationwide registry data, ensuring complete population coverage and eliminating selection bias. Prescription data include all community pharmacy dispensations, regardless of prescriber or sector.

Several limitations should be noted. Diagnoses made by private practising child psychiatrists are not captured in the Danish National Patient Registry, which leads to underestimation and potentially biases geographic comparisons. However, in 2024, 91% of capturable ADHD diagnoses made by child and adolescent psychiatrists were recorded in the hospital sector and are therefore present in our data [39]. Second, prevalence was defined as at least one prescription in 2024, which may overestimate active treatment rates, though relative geographic patterns are unlikely to change.

Waiting time was available only for hospital-based child and adolescent psychiatry. Current expected waiting times reported by private practising psychiatrists working under the public reimbursement system can be obtained [40], but historical data are lacking.

Specialist capacity data were available only for private practising psychiatrists working under the public reimbursement system. Information on public hospital-based capacity is available in the Health Workforce Mobility Register [41], which was not obtained for this study, and publicly accessible data on hospital capacity divided by region extend only until 2023 [42].

Statistical associations at the regional level were strongly limited by the small number of regions ($n = 5$), which reduced power and increased uncertainty around effect estimates. Furthermore, the outcome measures were assessed at municipal level ($n = 98$), whereas the supply-side indicators (waiting times and psychiatrist capacity) were only available at regional level ($n = 5$), a resolution mismatch that may partly explain the null associations observed for these factors.

Given the number of correlation and regression analyses performed across municipal, regional and sex-stratified indicators, the borderline association between socioeconomic index and medication use ($\rho = -0.20$, $p = 0.049$) should be interpreted cautiously, as we did not correct for multiple comparisons.

Additionally, the organisation of child and adolescent mental health services varies geographically, with some areas served by multiple units and others by only one. In the latter case, a single department can account for a large proportion of diagnoses in the region, potentially amplifying local practice effects. However, information on hospital catchment areas was not consistently available across all Danish regions, limiting our ability to examine whether variation clustered around individual departments.

Geographic attribution was based on municipality of residence in 2024, although ADHD diagnoses or treatment may have originated elsewhere following residential migration.

Finally, we lacked data on symptom severity, parental demand, local clinical practice norms and individual-level socioeconomic background, all of which may be important explanatory factors.

5 | Conclusion

In conclusion, large geographic differences in ADHD diagnoses and medication use persist in Denmark despite a uniform healthcare framework. These differences appear only weakly related to specialist capacity, waiting times and socioeconomic context, suggesting that variation might be driven by other factors. Reducing inequities will likely require a multifaceted approach that addresses both resource distribution and local practice variation, ensuring equitable and consistent pathways to diagnosis and treatment.

Acknowledgements

Jacob Harbo Andersen is acknowledged for the validation of the analytical code.

Funding

This work was supported by Novo Nordisk Fonden, 10.13039/501100009708, NNF24SA0093635; Det Obelske Familiefond, 10.13039/501100007290, 69348. Funding sources had no role in designing, planning, analysing, interpreting, or reviewing the manuscript.

Ethics Statement

According to Danish law, studies based solely on register data do not require ethical approval.

17427843, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/bcpt.70298 by Stony Brook University Libraries, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Individual-level data cannot be shared by the authors owing to Danish data protection regulations. De-identified data can be made available for authorised researchers after application with a relevant aim to Forskerservice at the Danish Health Data Authority.

References

1. M. Bliddal, E. B. Gram, H. Sonne, et al., "Trends and Incidence Rates of Neurodevelopmental Disorders in Danish Children and Adolescents 2000–2024," *European Child & Adolescent Psychiatry* 35 (2026): 1667–1675, https://doi.org/10.1007/s00787-026-02981-0.

2. T. Hofstad, O. Nyttingnes, I. Bjelland, and A. Mykletun, "Catchment Area Characteristics Do Not Account for Geographical Variation in ADHD Diagnoses," *European Child & Adolescent Psychiatry* 34 (2025): 3097–3103, https://doi.org/10.1007/s00787-025-02720-x.

3. T. Widding-Havneraas, S. Markussen, F. Elwert, et al., "Geographical Variation in ADHD: Do Diagnoses Reflect Symptom Levels?," *European Child & Adolescent Psychiatry* 32 (2023): 1795–1803, https://doi.org/10.1007/s00787-022-01996-7.

4. A. Zgodic, A. C. McLain, J. M. Eberth, A. Federico, J. Bradshaw, and K. Flory, "County-Level Prevalence Estimates of ADHD in Children in the United States," *Annals of Epidemiology* 79 (2023): 56–64, https://doi.org/10.1016/j.annepidem.2023.01.006.

5. J. Wennberg and A. Gittelsohn, "Small Area Variations in Health Care Delivery," *Science* 182 (1973): 1102–1108, https://doi.org/10.1126/science.182.4117.1102.

6. S. Urwin, L. Anselmi, Y.-S. Lau, and M. Sutton, "Are Geographic Variations in Secondary Hospital Expenditure Caused by Supply and Demand Factors? Evidence From Migration in England," *Social Science & Medicine* 383 (2025): 118439, https://doi.org/10.1016/j.socscimed.2025.118439.

7. J. E. Wennberg, *Tracking Medicine: A Researcher's Quest to Understand Health Care* (Oxford University Press, 2010).

8. K. B. Madsen, S. Antonsen, R. T. Wesselhøft, et al., "Geographic Variation of Recorded Neurodevelopmental Disorders in Children and Adults," *Danish Medical Journal* 71 (2024): 1–9, https://doi.org/10.61409/A02240144.

9. M. Schmidt, S. A. J. Schmidt, J. L. Sandegaard, et al., "The Danish National Patient Registry: A Review of Content, Data Quality, and Research Potential," *CLEP* 2015 (2015): 449–490, https://doi.org/10.2147/CLEP.S91125.

10. A. Pottegård, S. A. J. Schmidt, H. Wallach-Kildemoes, H. T. Sørensen, J. Hallas, and M. Schmidt, "Data Resource Profile: The Danish National Prescription Registry," *International Journal of Epidemiology* 46 (2017): 798–798f, https://doi.org/10.1093/ije/dyw213.

11. C. B. Pedersen, "The Danish Civil Registration System," *Scandinavian Journal of Public Health* 39 (2011): 22–25, https://doi.org/10.1177/1403494810387965.

12. Økonomi- og Indenrigsministeriet, *The Municipal Indicators* (in Danish) De Kommunale Nøgletal (Økonomi- og Indenrigsministeriet, 2025), https://www.noegletal.dk/noegletal/ntStart.html. Accessed 1 July 2025

13. Sundhedsdatastyrelsen, *Waiting Times for Hospital Treatment* (in Danish) (Sundhedsdatabanken, 2025), https://sundhedsdatabank.dk/behandling-og-pleje/ventetid-menu/ventetid-til-behandling Accessed 8 May 2026.

14. Sundhedsdatastyrelsen, *Other Providers in Private Practice* (in Danish) (eSundhed.dk, 2025), https://www.esundhed.dk/Emner/Beskaeftigede-i-sundhedsvaesnet/Andre-ydere-i-praksis Accessed 1 July 2025.

15. Sundhedsdatastyrelsen, *The Legal Right to Diagnostic Assessment* (in Danish). (Sundhedsdatabanken, 2026), https://sundhedsdatabank.dk/behandling-og-pleje/udredningsret-menu/udredningsret-i-boerne-og-ungdomspsykiatrien Accessed 29 June 2026.

16. M. Schmidt, S. A. J. Schmidt, K. Adelborg, et al., "The Danish Health Care System and Epidemiological Research: From Health Care Contacts to Database Records," *Clinical Epidemiology* 11 (2019): 563–591, https://doi.org/10.2147/CLEP.S179083.

17. H. O. Birk, K. Vrangelbæk, A. Rudkjøbing, et al., *Denmark: Health System Review* (WHO Regional Office for Europe, 2024).

18. Danmarks Statistik, *The Bank of Statistic's* (in Danish) (Statistikbanken, 2025), https://www.statistikbanken.dk/statbank5a/default.asp?w=1920 Accessed 21 July 2025.

19. Sundhedsstyrelsen, *Specialty Guideline for Child and Adolescent Psychiatry* (in Danish) (Sundhedsstyrelsen, 2023).

20. Psykiatrien i Region Midtjylland, *Central Psychiatric Referral Unit* (in Danish) (Psykiatrien i Region Midtjylland - til fagpersoner, 2025), https://www.fagperson.psykiatrien.rm.dk/henvisning-og-visitation/psykiatriens-centrale-visitation/ Accessed 28 Sept 2025.

21. Sundhedsstyrelsen, *Analysis of Waiting Times for Practising Psychiatrists and Child and Adolescent Psychiatrists* (in Danish) (Sundhedsstyrelsen, 2024), https://www.sst.dk/-/media/Udgivelser/2025/Psykiatri/Analyse-af-ventetider-til-praktiserende-psykiatere-og-boerne--og-ungdomspsykiatere.ashx?sc_lang=da&hash=253FA3CDFC37F6D6E8CC6F61E4D565C4. Accessed 28 Sept 2025

22. American Psychiatric Association, DSM-5 Task Force, *Diagnostic and Statistical Manual of Mental Disorders: DSM-5* (American Psychiatric Publishing, Inc., 2013).

23. S. Dalsgaard, E. Thorsteinsson, B. B. Trabjerg, et al., "Incidence Rates and Cumulative Incidences of the Full Spectrum of Diagnosed Mental Disorders in Childhood and Adolescence," *JAMA Psychiatry* 77 (2020): 155–164, https://doi.org/10.1001/jamapsychiatry.2019.3523.

24. Sundhedsdatastyrelsen, *Medstat.dk* (MEDSTAT.DK, 2025), https://medstat.dk/ Accessed 13 Oct 2025.

25. World Health Organization, *ICD-10: International Statistical Classification of Diseases and Related Health Problems : Tenth Revision* (World Health Organization, 2004).

26. P. Tveden-Nyborg, T. K. Bergmann, N. Jessen, U. Simonsen, and J. Lykkesfeldt, "BCPT 2026 Policy for Experimental and Clinical Studies," *Basic & Clinical Pharmacology & Toxicology* 137 (2025): e70159, https://doi.org/10.1111/bcpt.70159.

27. L. Pérez-Crespo, J. Canals-Sans, E. Suades-González, and M. Guxens, "Temporal Trends and Geographical Variability of the Prevalence and Incidence of Attention Deficit/Hyperactivity Disorder Diagnoses Among Children in Catalonia, Spain," *Scientific Reports* 10 (2020): 6397, https://doi.org/10.1038/s41598-020-63342-8.

28. V. Prasad, J. West, D. Kendrick, and K. Sayal, "Attention-Deficit/Hyperactivity Disorder: Variation by Socioeconomic Deprivation," *Archives of Disease in Childhood* 104 (2019): 802–805, https://doi.org/10.1136/archdischild-2017-314470.

29. A. E. Russell, T. Ford, R. Williams, and G. Russell, "The Association Between Socioeconomic Disadvantage and Attention Deficit/Hyperactivity Disorder (ADHD): A Systematic Review," *Child Psychiatry and Human Development* 47 (2016): 440–458, https://doi.org/10.1007/s10578-015-0578-3.

30. T. E. Froehlich, B. P. Lanphear, J. N. Epstein, W. J. Barbaresi, S. K. Katusic, and R. S. Kahn, "Prevalence, Recognition, and Treatment of Attention-Deficit/Hyperactivity Disorder in a National Sample of US

17427843, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/bcpt.70298 by Stony Brook University Libraries, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Children," *Archives of Pediatrics & Adolescent Medicine* 161 (2007): 857–864, https://doi.org/10.1001/archpedi.161.9.857.

31. K. B. Madsen, A. K. Ersbøll, J. Olsen, E. Parner, and C. Obel, "Geographic Analysis of the Variation in the Incidence of ADHD in a Country With Free Access to Healthcare: A Danish Cohort Study," *International Journal of Health Geographics* 14 (2015): 24, https://doi.org/10.1186/s12942-015-0018-4.

32. H. Wallach-Kildemoes, A. M. Skovgaard, K. Thielen, A. Pottegård, and L. H. Mortensen, "Social Adversity and Regional Differences in Prescribing of ADHD Medication for School-Age Children," *Journal of Developmental & Behavioral Pediatrics* 36 (2015): 330–341, https://doi.org/10.1097/DBP.0000000000000170.

33. J.-F. Levesque, M. F. Harris, and G. Russell, "Patient-Centred Access to Health Care: Conceptualising Access at the Interface of Health Systems and Populations," *International Journal for Equity in Health* 12 (2013): 18, https://doi.org/10.1186/1475-9276-12-18.

34. B. D. Fulton, R. M. Scheffler, S. P. Hinshaw, et al., "National Variation of ADHD Diagnostic Prevalence and Medication Use: Health Care Providers and Education Policies," *Psychiatric Services* 60 (2009): 1075–1083, https://doi.org/10.1176/ps.2009.60.8.1075.

35. I. Lyhmann, T. Widding-Havneraas, H. D. Zachrisson, et al., "Variation in Attitudes Toward Diagnosis and Medication of ADHD: A Survey Among Clinicians in the Norwegian Child and Adolescent Mental Health Services," *European Child & Adolescent Psychiatry* 32 (2023): 2557–2567, https://doi.org/10.1007/s00787-022-02110-7.

36. S. Dalsgaard, H. S. Nielsen, and M. Simonsen, "Consequences of ADHD Medication Use for Children's Outcomes," *Journal of Health Economics* 37 (2014): 137–151, https://doi.org/10.1016/j.jhealeco.2014.05.005.

37. T. Tegtmejer and R. Säljö, "Paradoxes of Using Psychiatric Diagnoses as Organising Principles for Special Educational Support," *European Journal of Special Needs Education* 39 (2024): 519–533, https://doi.org/10.1080/08856257.2023.2242029.

38. K.-Y. Liu, M. King, and P. S. Bearman, "Social Influence and the Autism Epidemic," *AJS* 115 (2010): 1387–1434, https://doi.org/10.1086/651448.

39. Sundhedsdatastyrelsen, [*The Visualization of 'Diagnosis Data From Specialist Practice'*] (in Danish) (Sundhedsdatastyrelsen, 2025) https://powerbi.esundhed.dk/Reports/powerbi/esundhed_aaben/emner/RSP/a_rsp_diagnose?rs:embed=true. Accessed 16 Dec 2025

40. Danske Regioner, *Find a Provider*] (in Danish) (sundhed.dk, 2025), https://www.sundhed.dk/borger/guides/find-behandler/?RegionId=0&InformationsUnderkategori=Psykiater&Informationskategori=Speciall%C3%A6ge,Speciall%C3%A6ge&WaitTime=false Accessed 28 Sept 2025.

41. Sundhedsdatastyrelsen, *The Health Workforce Mobility Register* (in Danish) (eSundhed.dk, 2025), https://www.esundhed.dk/Registre/Bevaegelsesregisteret Accessed 28 Sept 2025.

42. Sundhedsdatastyrelsen, *The Health-Educated Workforce* (in Danish) (Sundhedsdatabanken, 2024), https://sundhedsdatabank.dk/sundhedsvaesenets-rammer/uddannede-sundhedspersoner-i-arbejdsstyrken Accessed 28 Sept 2025.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Number of inhabitants aged 4–17 years in Denmark, 2024, at national, regional and municipal levels. The denominator was estimated as the number of individuals alive on 1 July 2024, within the age range for each region and municipality. **Table S2:** Prevalence of ADHD medication use and ADHD diagnoses among children and adolescents aged 4–17 years in Denmark, 2024. Estimates are shown per 1000 inhabitants at national, regional and municipal levels. **Table S3:** Prevalence of ADHD medication use among children and adolescents aged 4–17 years in Denmark, 2024, stratified by age group. Estimates are shown per 1000 inhabitants at national, regional and municipal levels. **Table S4:** Prevalence of ADHD diagnoses among children and adolescents aged 4–17 years in Denmark, 2024, stratified by age group. Estimates are shown per 1000 inhabitants at national, regional and municipal levels. **Table S5:** Prevalence of ADHD medication use among children and adolescents aged 4–17 years in Denmark, 2024, stratified by sex. Estimates are shown per 1000 inhabitants at national, regional and municipal levels. **Table S6:** Prevalence of ADHD diagnoses among children and adolescents aged 4–17 years in Denmark, 2024, stratified by sex. Estimates are shown per 1000 inhabitants at national, regional and municipal levels. **Table S7:** Regional and municipal indicators in Denmark, 2024. Indicators include municipal socioeconomic index (values > 1 indicate higher-than-average needs), average regional waiting times in public child and adolescent psychiatry (days) and regional capacity of private practicing child and adolescent psychiatrists with a provider number (full-time equivalents per 100 000 inhabitants aged 4–17 years). **Table S8:** Spearman's rank correlation coefficients between prevalence of ADHD medication use or ADHD diagnoses and regional/municipal indicators in Denmark, 2024. Results are shown for children and adolescents aged 4–17 years overall and stratified by sex. **Table S9:** Results from linear regression analyses of the association between prevalence of ADHD medication use or ADHD diagnoses and regional/municipal indicators in Denmark, 2024. Analyses are presented for children and adolescents aged 4–17 years overall and stratified by sex. Municipal-level analyses are population-weighted, whereas regional-level analyses are unweighted. **Table S10:**. Results from linear regression analyses testing for interaction between sex and socioeconomic index in their association with prevalence of ADHD medication use and ADHD diagnoses among children and adolescents aged 4–17 years in Denmark, 2024. Estimates represent the interaction term (sex × socioeconomic index), with boys as the reference category. **Figure S1:** Number of inhabitants aged 4–17 years in Denmark, 2024, at the regional level. The numbers were estimated as the number of individuals aged 4–17 years within each region alive on 1 July 2024. **Figure S2:** Municipal prevalence of ADHD medication use per 1000 inhabitants among children and adolescents aged 4–17 years in Denmark, 2024, stratified by age group. Corresponding numerical values are provided in Table S3. Grey areas indicate municipalities with non-reportable counts ($N < 10$). **Figure S3:** Municipal prevalence of ADHD diagnoses per 1000 inhabitants among children and adolescents aged 4–17 years in Denmark, 2024, stratified by age group. Corresponding numerical values are provided in Table S4. Grey areas indicate municipalities with non-reportable counts ($N < 10$). **Figure S4:** Municipal prevalence of ADHD medication use per 1000 inhabitants among children and adolescents aged 4–17 years in Denmark, 2024, stratified by sex. Corresponding numerical values are provided in Table S5. Grey areas indicate municipalities with non-reportable counts ($N < 10$). **Figure S5:** Municipal prevalence of ADHD diagnoses per 1000 inhabitants among children and adolescents aged 4–17 years in Denmark, 2024, stratified by sex. Corresponding numerical values are provided in Table S6. Grey areas indicate municipalities with non-reportable counts ($N < 10$).

17427843, 2026, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/bcpt.70298 by Stony Brook University Libraries, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License